



Read the Turtle Code

Ontario Math C3.2 — read code and predict

Name: _____

Date: _____

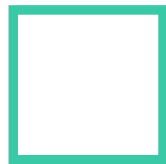
★ I can read a turtle loop and predict what Bit draws.

Picture it first

Good coders read code and picture what it does BEFORE running it. For a turtle loop, look for two clues: is there a turn (a shape) or no turn (a straight line), and how many sides (the range number).

Read each loop below and predict Bit's drawing. Then you can run it to check.

read the loop → picture what Bit draws



with a turn

→ a shape



no turn

→ a straight line

A turn makes a shape; no turn makes a straight line.

Loop A — read it

```
for i in range(4):  
    bit.forward(30)  
    bit.right(90)
```

PYTHON

1 Read Loop A

Loop A has a turn (right 90) and range(4). What shape does Bit draw, and how many sides? _____

2 Read another loop

Loop B is: for i in range(3): bit.forward(40); bit.left(120). What shape does Bit draw, and how many sides?

3 Predict, then run

Loop C is: for i in range(5): bit.forward(20) (no turn). PREDICT what Bit draws and how long it is in total. Then run it.

4 Challenge

Loop A draws a square with range(4), right 90. After the loop finishes, WHERE is Bit — back at the start, or somewhere else? How do you know?

Big idea

You can read a loop and picture the drawing: a turn means a shape, no turn means a line, and the range number tells you how many sides.